

Phase 1.1.3

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Photos of the Low-Fidelity Prototype

Home Page	Transaction/ Categories Page	Goals/ Spending Reports Page

Prioritized List of System Requirements

1. Add and Categorize Transactions:
 - a. Users will be able to manually enter their income and expenses and then assign them to specific categories (e.g., groceries, rent, entertainment).
2. Set Budget Limits by Category:
 - a. Users will be able to establish spending cap limits per category, and receive notifications when either nearing or exceeding those limits.
3. Set Upcoming Payment Reminders:
 - a. Users will be able to input recurring bills or payments (e.g., rent, subscriptions, utilities, mortgage) and receive a set number of reminders before the payments are due.
4. Set and Track Savings Goals:

- a. Users will be able to create specific savings goals (e.g., vacation, emergency fund), define target amounts and deadlines, and track the progress of these goals.
- 5. View Spending Reports and Trends:
 - a. Users will be able to generate and view reports that show spending trends and progress towards goals set.

Primary and Secondary Features

Primary Features

1. Transaction Entry
 - a. Users click “Add Transaction” to open a pop-up form where they choose income or expense, select a category from a dropdown, enter the amount, and optionally add a note before saving it.
2. Budget Limits per Category
 - a. Users set a spending limit for each category using dropdowns and input fields.
3. Upcoming Payment Reminders
 - a. Users enter upcoming bills by filling out a form with the amount, due date, and how often it repeats, and a reminder appears on the home screen a few days before the payment is due.

Secondary Features

1. Savings Goals
 - a. Users create a goal by entering a name, target amount, and deadline, and a simple progress bar shows how close they are to reaching it based on their saved amounts.
2. Spending Reports
 - a. Users can view reports that show how much they’ve spent or earned over time, with visual breakdowns by category and time period to help spot trends and habits.

Who Did What: Phase 1

Phase 1.1.1

In this phase, everyone helped build the foundation of the app. Melanie wrote a summary explaining what the app is and what it does. Aditi added the primary goals and described the overall audience the app is meant for. Samantha contributed by writing the secondary goals. Aidana worked on the ethical considerations and potential issues section.

Phase 1.1.2

The team created task scenarios and user personas to better understand how different users might interact with the app. Melanie wrote the first two task scenarios: College Student and Single Mother. Aditi added two more: High School Graduate and PhD Student with Loans. Aidana wrote two additional ones: Industry Professional and Retiree. Samantha created the user personas for Tony Renaldo and Emilia Bouilloire.

Phase 1.1.3

Samantha created a 3-page low-fidelity prototype on paper. She wrote the style guide, including the color scheme (which the team chose together using the Coolors website), and selected the font and iconography on her own, and then created a photorealistic prototype of the home page using Canva. Melanie wrote the prioritized list of system requirements and updated the task scenarios from phase 1.1.2 based on the feedback we received. Aditi wrote the list of primary and secondary features, and also wrote this write-up outlining who did what for the entirety of phase 1. Aidana wrote the discussion based on the user feedback we received.

Phase 1.2

For Phase 1.2, Aditi worked on the profile HTML page and the colors HTML page. She also added the navigation bar with icons to each page and created the css file to style the color scheme and nav bar. Aidana worked on the text HTML page and the to-do HTML page. Melanie created the LR HTML page. Samantha made the choices HTML page and put together the PDF submission, including screenshots of all six pages.

Prototype User Feedback

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User A is a fourth-year college student, studying psychology and criminology. She works for the college she attends and receives free housing and a one-time stipend at the beginning of the semester. She doesn't pay rent. She wants to use the budgeting app to help her save up for an upcoming trip.

User B is also a fourth-year college student, studying biology. He does not have a job, but instead receives money from his parents every month. He wants to use the budgeting app to help him keep track of this allowance and make sure it's enough to cover groceries and all other necessary expenses. He lives off-campus and needs to pay rent every month.

User C is a second-year college student, studying accounting. She has a part-time job and is paid hourly. She does not pay rent because she lives in university housing. She loves shopping and sometimes overspends. She wants to use the budgeting app to help her spend less on certain categories.

Home Screen

User B liked the large notification banner at the top of the home screen with the deadline reminders because it immediately caught his attention, fulfilling its purpose. He mentioned that it might make more sense for the home button on the navbar to be in the center. User C thought it was unclear what certain buttons on the navbar led to. She mentioned supplementing the icons with text like "Transactions" and "Spending" to make it more obvious. She also thought there were too many elements on the home screen, and some of the features were already shown on the other pages.

Transaction Screen

User C liked how the "Add New Transaction" part had customizable entry choices. She pointed out how the name and amount could be typed manually by the user, but the category is a dropdown. This results in a quicker input time. She mentioned that the transaction category should also be shown in the transaction history area. User A suggested that the items in the transaction history could be color-coded by category when the medium-fidelity prototype is developed.

Spending Screen

Users A and B liked how there is an option to see how much was spent by day, month, and year. User C pointed out that it would also be a good idea to have a week option, because day is too short but month is too long, so week is a better timeframe. User A liked how the spending/report screen wasn't just a block of text, but had various types of graphs, bars, and other data

visualizations. User C liked how the goal was represented by a progress bar because it's easy to understand and serves as a good visual motivator.

Overall App Design

User B liked the presence of the notification banner but noticed that there wasn't a button/option that allowed one to edit or set these reminders. He also said that both transaction and spending had to do with money leaving, and there was no page specifically for income or money being saved. He believes that it would be more intuitive to have a home page, a transactions/spending tab in one, and then a separate income/savings tab.

Fixed List of User Personas and Task Scenarios (Based on Feedback)

Task Scenario 1: College Student

A college student, Jordan, just received their monthly paycheck from their part-time campus job at SORC in the Stamp Student Union. She wants to make sure she can pay rent, cover groceries, and still save \$100 toward a spring break trip with friends.

Source of scenario: The source of this scenario is from a friend here at UMD that works part-time at SORC in the Stamp Student Union. We interviewed her to hear about how she splits her earnings each month, and what tools she'd like to see in the application.

Why this is representative: This scenario is representative of a college student that works part-time to support their bills and possibly tuition. It's representative of a larger population of students that balance limited income with multiple fixed expenses (e.g., rent) and short-term goals (e.g., spring break trip). This limited income partnered with short-term financial responsibility makes this population of students an ideal demographic for our budgeting application.

Task Scenario 2: Single Mother

Maria is a 27-year-old single mother to a 5-year-old son working two part-time jobs as a barista and an Amazon delivery driver. Her hours vary greatly week to week, so her income is therefore unpredictable. Her expenses include rent, utilities, groceries, and public transportation costs. There are also other less frequent expenses like doctor visits and school fees for her son. Maria often struggles to plan ahead financially, and often borrows money from friends or delays paying her bills.

Source: This scenario is based on someone I met briefly while volunteering at the D.C. Central Kitchen. We chatted about her work-life balance, and she gave me some insight into a few of the difficulties she was facing.

Why this is representative: This scenario is representative of a young, single parent who may be more reliant on each paycheck to pay off expenses. It's representative of a larger population of individuals with unstable income who prioritize flexibility and short term budgeting. Incorporating scenarios like this one ensures that the app is inclusive to those in all types of financial situations.

Task Scenario 3: High School Graduate

Pam is a recent high school graduate and tennis player who is saving up to buy her own car before she goes off to college in September. She has been working as a receptionist at a local hair salon this summer. Since she still lives at home, she has no rent, grocery, or other living expenses, as she is still covered by her parents for the time being. All of her earnings are counted towards her savings for the car, meaning she does not have to categorize any of her expenses.

Source of scenario: This scenario is based on a situation my cousin went through a few years ago. I recently chatted with her about it, and she mentioned how helpful an app like this would have been at the time to track her savings and plan her work schedule more efficiently.

Why this is representative: This scenario represents a common situation for older teens and young adults who are financially dependent on their families, but beginning to manage their own personal expenses. Many high school graduates take on summer jobs to save up before college. Their financial profile is unique because of their minimal expenses, irregular income, and a strong focus on a single savings goal. This makes them a key demographic for a budgeting app designed to be simple and organized.

Task Scenario 4: PhD Student with Loans

Carlos just started a PhD program in psychology at UC San Diego after graduating from college. His fees are fully covered through a fellowship, and he received a monthly stipend for his work as a research assistant in the Development Psychology department. Although he does not have to worry about tuition, Carlos needs to cover his rent, groceries, transportation, and start repaying his undergraduate student loans.

Source of scenario: This scenario is based on a friend who is currently a senior in college and is applying for PhD programs. He anticipates having a stipend and getting her fees covered through

a fellowship, but knows that she would benefit from using an app to make sure she covers her regular expenses and work towards her debt.

Why this is representative: Many young adults who want to pursue multiple degrees face similar situations. Student loan debt can be a huge financial burden unless people plan effectively. A budgeting app like this can help them manage limited income, track expenses, and make steady progress towards their debt.

Task Scenario 5: Industry Professional

Thomas is a 39-year old project manager at a consulting company based in Virginia. He is married, has one child (a 14-year old), and owns a spacious three-bedroom home. He is still paying off their mortgage for that, which has about twelve years remaining. They also own a car, but the loan for that has already been paid off. His wife is a stay-at-home mom, so Thomas takes care of all of the essential expenses. In addition to these monthly bills, he makes sure to contribute regularly to his kid's college savings fund.

Source of scenario: The source of this scenario is a combination of persona brainstorming and the experiences of a family friend.

Why this is representative: Thomas's situation is a common reality for many industry professionals in America. He has a stable career and lives a comfortable middle class lifestyle, but is still very financially conscious. A budgeting app like this can be a great tool for both short-term and long-term planning.

Task Scenario 6: Retiree

Ethel is a retired schoolteacher who lives by herself in a small apartment that she bought. She used to be married, but her husband passed away a long time ago. Since she owns the apartment, her largest household expenses are utilities and taxes. Other than that, her main financial concerns are health-related costs, like recurrent prescriptions and medical appointments. She receives Social Security and a pension from her previous employer, which is barely enough to cover everything. Ethel doesn't have the best memory. She owns an iPhone, so she's familiar with apps, but not a tech-expert.

Source of scenario: This scenario is based on my grandmother. I stay with her every summer, so I know a lot about her monthly income, spending habits, and financial burdens.

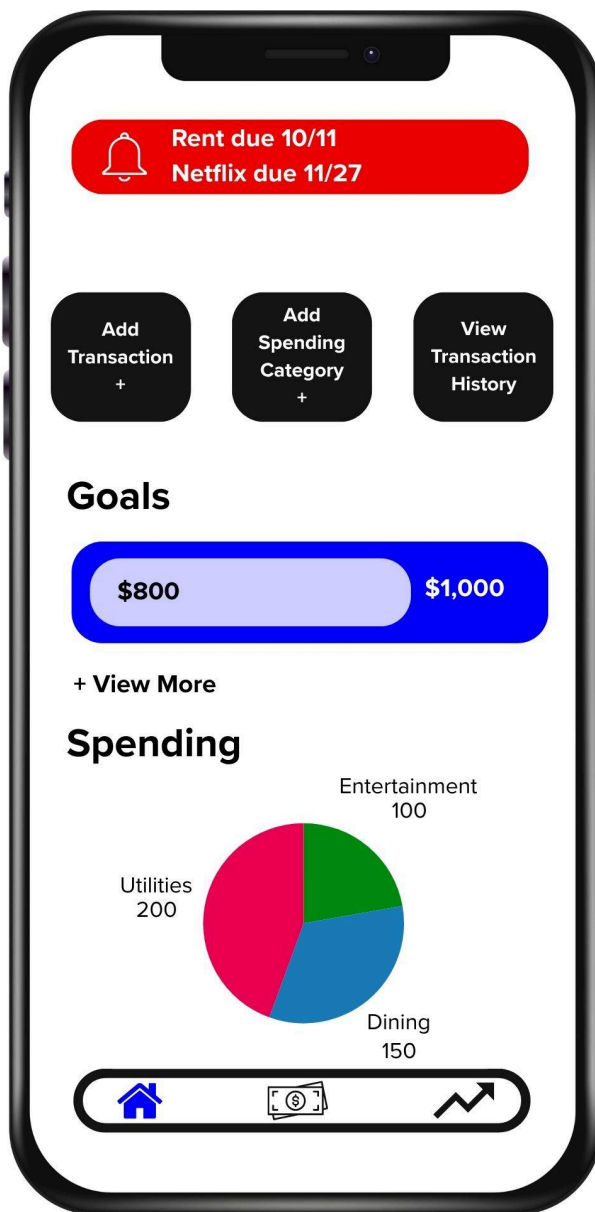
Why this is representative: This scenario is representative of many older retirees with a fixed, low income. Like Ethel, they are often not very tech-savvy and may have a poor memory. A

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budgeting app with a simple UI could be an extremely useful tool for such people. It can help manage essential bills and reduce stress with remembering deadlines.

Homepage Mockup



Style Guide

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Font: Proxima Nova (Regular, Semibold, Bold)

Font Size: 48 pt for headers (Goals, Spending), 30 pt for dropdown labels, 26 pt for buttons and text

Color Scheme:

- Primary: #0000FF for main actions, highlights, and progress fills
- Accent: #EB0052, #008A10, #197BB8 for charts
- Alert/Destructive: #EE0000 for banners and warnings
- Backgrounds: #FFFFFF (main), #CDCDFF (light fill)
- Text: #000000 for headings/body/borders

Iconography:

Home (homepage), Money (transactions), Line Graph (spending reports), Bell (reminders).

Icons are 24–28 px, active icons use #0000FF, inactive icons use black (#000000), and icons on colored surfaces render in white